

# **ANALYSIS REPORT**

## **RECURRENCE RISK PREDICTION FOR THYROID CANCER ACCORDING TO ATA 2025 GUIDELINES**

Retrospective Study, 114 patients  
Study period: 2024

*Data encoded according to ATA 2025 standards*

## 1. Study Sample Characteristics

Total patients: 114. Recurrence events: 7 (6.1%).

### 1.1. ATA 2025 Risk Stratification

| Risk Level   | N   | %     | Recurrence   |
|--------------|-----|-------|--------------|
| Low          | 27  | 23.7% | 0/27 (0.0%)  |
| Intermediate | 61  | 53.5% | 5/61 (8.2%)  |
| High         | 26  | 22.8% | 2/26 (7.7%)  |
| Total        | 114 | 100%  | 7/114 (6.1%) |

### 1.2. Descriptive Statistics of Study Variables

| Variable                 | N (n=114) | %     |
|--------------------------|-----------|-------|
| Multifocal (>1 nodule)   | 31        | 27.2% |
| Bilateral involvement    | 32        | 28.1% |
| Tumor size >1cm          | 21        | 18.4% |
| Tumor size >4cm          | 1         | 0.9%  |
| T stage T3+              | 28        | 24.6% |
| LN metastasis (N+)       | 59        | 51.8% |
| Lateral neck meta. (N1b) | 8         | 7.0%  |
| Central LN >=5           | 16        | 14.0% |
| Lateral neck LN (+)      | 21        | 18.4% |
| BRAF V600E               | 45        | 39.5% |
| Other BRAF               | 2         | 1.8%  |
| TERT mutation            | 6         | 5.3%  |
| TP53 mutation            | 2         | 1.8%  |
| Any mutation             | 51        | 44.7% |
| Co-mutation (>=2 genes)  | 4         | 3.5%  |
| BRAF + TERT              | 3         | 2.6%  |
| BRAF + TP53              | 0         | 0.0%  |

## 2. Univariate Analysis Results

### 2.1. Univariate Logistic Regression

The table below presents univariate logistic regression results for each variable against recurrence.

| Variable                 | N   | OR     | 95% CI         | p-value |
|--------------------------|-----|--------|----------------|---------|
| Multifocal (>1 nodule)   | 114 | 2.116  | 0.446 - 10.048 | 0.346   |
| Bilateral involvement    | 114 | 2.017  | 0.425 - 9.566  | 0.377   |
| Tumor size >1cm          | 114 | 0.725  | 0.083 - 6.363  | 0.772   |
| T stage T3+              | 114 | 0.494  | 0.057 - 4.288  | 0.522   |
| LN metastasis (N+)       | 114 | 6.113  | 0.712 - 52.515 | 0.099   |
| Lateral neck meta. (N1b) | 114 | 2.381  | 0.251 - 22.622 | 0.450   |
| Central LN >=5           | 114 | 2.657  | 0.469 - 15.039 | 0.269   |
| Lateral neck LN (+)      | 114 | 0.725  | 0.083 - 6.363  | 0.772   |
| BRAF V600E               | 114 | 1.161  | 0.247 - 5.449  | 0.850   |
| BRAF (any)               | 114 | 1.074  | 0.229 - 5.038  | 0.928   |
| TERT mutation            | 114 | <0.001 | --             | 1.000   |
| TP53 mutation            | 114 | <0.001 | --             | 1.000   |
| Co-mutation (>=2)        | 114 | <0.001 | --             | 1.000   |
| BRAF + TERT              | 114 | <0.001 | --             | 0.999   |

*Note: TERT, TP53, co-mutation and BRAF+TERT have non-estimable OR (complete separation) due to zero recurrence events in mutation groups. OR <0.001 indicates complete separation and has no real statistical meaning.*

### 2.2. Key Observations from Univariate Analysis

- Lymph node metastasis (N+) was the strongest predictor of recurrence (OR=6.11; p=0.099), approaching but not reaching statistical significance at the 0.05 level.
- Multifocality (OR=2.12), bilaterality (OR=2.02), and central LN >=5 (OR=2.66) showed a trend toward increased recurrence risk.
- BRAF V600E showed OR=1.16, not statistically significant (p=0.85).
- TERT and TP53 had very few mutations in the sample (6 and 2 patients) with zero recurrence events, causing complete separation.

### 3. Multivariate Analysis Results

Multivariate logistic regression with 4 different models.

#### 3.1. Model 1 - Clinical variables (N=114, events=7)

| Variable  | OR    | 95% CI         | p-value |
|-----------|-------|----------------|---------|
| Size >1cm | 0.528 | 0.058 - 4.797  | 0.570   |
| T3+       | 0.415 | 0.047 - 3.701  | 0.431   |
| N+        | 7.027 | 0.804 - 61.429 | 0.078   |

#### 3.2. Model 2 - Genetic variables (N=114, events=7)

| Variable   | OR     | 95% CI        | p-value |
|------------|--------|---------------|---------|
| BRAF V600E | 1.144  | 0.243 - 5.384 | 0.865   |
| TERT       | <0.001 | --            | 1.000   |

#### 3.3. Model 3 - Combined clinical + genetic (N=114, events=7)

| Variable   | OR     | 95% CI         | p-value |
|------------|--------|----------------|---------|
| Size >1cm  | 0.511  | 0.056 - 4.671  | 0.552   |
| T3+        | 0.393  | 0.041 - 3.798  | 0.420   |
| N+         | 6.933  | 0.786 - 61.193 | 0.081   |
| BRAF V600E | 1.070  | 0.206 - 5.569  | 0.936   |
| TERT       | <0.001 | --             | 1.000   |

#### 3.4. Model 4 - N+ and co-mutation (N=114, events=7)

| Variable    | OR     | 95% CI         | p-value |
|-------------|--------|----------------|---------|
| N+          | 5.885  | 0.684 - 50.617 | 0.107   |
| Co-mutation | <0.001 | --             | 1.000   |

## 4. Key Findings and Conclusions

### - Recurrence Rate

The overall recurrence rate was 6.1% (7/114 patients), consistent with studies on papillary thyroid cancer patients receiving curative treatment.

### - ATA 2025 Stratification

ATA 2025 risk distribution: low 23.7%, intermediate 53.5%, high 22.8%. Recurrence rates increased from low (0%) to intermediate (8.2%) and high (7.7%), suggesting the stratification has discriminatory value.

### - Lymph Node Metastasis (N+)

N+ was the most important independent predictor, consistently identified in both univariate (OR=6.11; p=0.099) and multivariate analysis (OR=7.03; p=0.078). This was the strongest trend even though statistical significance was not reached due to small sample size.

### - Multifocality and Bilaterality

Multifocal (OR=2.12) and bilateral (OR=2.02) tumors showed a trend toward increased recurrence risk, consistent with ATA guidelines for intermediate-risk classification.

### - BRAF V600E

BRAF V600E mutation rate was 39.5% but showed no significant association with recurrence in this study (OR=1.16; p=0.85).

### - Rare Mutations

TERT (5.3%), TP53 (1.8%), and co-mutation (3.5%) had low frequencies. Complete separation occurred due to zero recurrence events in mutation groups, reflecting the small sample size and rare events.

### - Limitations

The limited number of recurrence events (n=7) substantially reduced statistical power, producing wide confidence intervals. Larger studies are needed to validate these associations, particularly for rare genetic variables. Survival analysis (Cox regression) may be more informative with longitudinal follow-up data.

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*Data encoded from Q3 source file and genetic results file. Variables coded according to ATA 2025 standards. Encoded data file: DATA\_MAHOA\_ATA2025.xlsx*